

PHT 100: ENGINEERING PHYSICS A

Max. Mark: 100

Duration: 3 hours

ANSWER KEY

PART A

Answer all questions. Each question carries 2 marks.

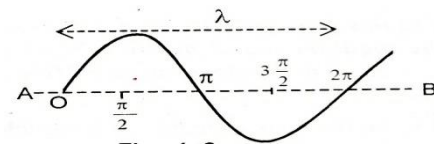
1. The amplitude of the forced oscillations at resonance is called resonant amplitude *when the frequency of the applied force is equal to the natural frequency of the oscillation.*

Amplitude at Resonance $A_{max} = \frac{f_0}{2\gamma\omega_f}$

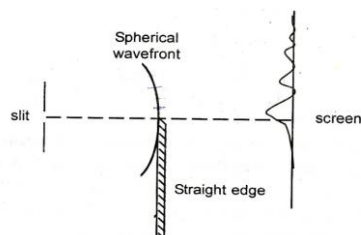
eg sonometer. resonance column

2. Path difference is the Difference in path between two waves and phase difference is the difference in phase between the two light waves.

Phase difference $= 2\pi/\lambda \times$ path difference



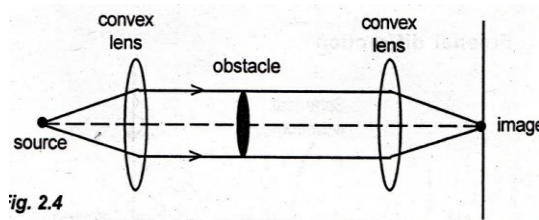
3. In air wedge the fringes are straight because thickness of air film is same along a line parallel to the line of contact of the glass plate. So the interference pattern are parallel and straight to the line of contact. Each fringe is the locus of all points where the thickness of air film has a constant value. But in newtons ring fringes are circular because the thickness of air film is same along a circle. If a point appears dark, all the points along a circle through this point are dark since. So we get a dark ring.
4. FRESNEL DIFFRACTION - Either the source of light or the screen or both are at finite distance from the obstacle causing diffraction. Wavefront falling on the obstacle is spherical. Lenses are not used.



Example :Diffraction at a straight edge

FRAUNHOFER DIFFRACTION

The source of light and the screen are at infinite distance with respect to the obstacle causing diffraction. Wavefront falling on the obstacle is plane. Lenses are used



Example: Diffraction at a grating

5. Classically

If $E < V_0$ all the particle will be reflected at the barrier

If $E > V_0$ all will be transmitted in to the region III

Quantum mechanically

If $E < V_0$ there will be a finite probability for the particle to be transmitted in to the region III

This phenomenon of tunnelling through the barrier is known as quantum mechanical tunnelling or barrier penetration

two electronic devices based on this phenomenon are tunnel diode and Zener diode

6. Quantum confinement is the restricted motion of randomly moving electron in specific energy levels when the dimensions of a material approach the de Broglie wavelength of electron. The quantum confinement effects is observed when the size of the particle is too small to be comparable to the de-broglie wavelength of the electron . Bulk material have continuous energy levels. In the case of nano range, the energy level become discrete. As the size reduces energy level changes from continuous to discontinuous. This increases the band gap energy. Motion of the electron become restricted.
7. Magnetic flux density (B) is defined as the total number of magnetic field lines passing perpendicularly through unit area. The flux density is expressed in Weber per sq. meter or Tesla. Magnetic field strength, also called magnetic intensity or magnetic field intensity (H), the part of the magnetic field in a material that arises from an external current. $B = \mu H$ where μ is the permeability of the material.
8. Conduction current is produced due to rate of change of charges. But the displacement current is produced due to rate of change of Electric Field.
Conduction current is zero in perfect vacuum but displacement current has a finite value in vacuum.
Conduction current is in phase with electric field where as the Displacement current leads the electric field by $\pi/2$ radians
9. According to this theory,
superconductivity is mainly due to the interaction of phonons and electrons. When an electron is advancing through a path in between 2 sets of positive ions in the lattice, electron attracts nearby positive ions and lattice gets distorted. This distortion produces a region of increased positive charge density and is called phonon. If a second electron approaches this region, this electron also attracted by the positive

region. Thus these 2 electrons are held together by force of attraction by phonon. Thus 2 electrons are coupled together at low temperature and this pair is called cooper pair.

10. Photo diode is a silicon or germanium p-n junction working under reverse biased condition. Reverse current increases when light falls on the junction. When a small reverse voltage is applied a very small current flows through the diode. This is due to minority charge carriers. At the same time majority charge carriers are prevented from crossing the junction. The current flowing through the reverse biased photodiode when no light is incident on the junction is called 'dark current'. The corresponding resistance is called 'dark resistance'

$$\text{Dark resistance} = \frac{V_R}{I_d}$$

where V_R is reverse voltage and I_d the dark current.



When light falls on the junction the energy of the photons produce more electron-hole pairs. The number of new electron-hole pairs is proportional to the number of incident photons

PART B

11(a) DAMPED HARMONIC OSCILLATION

A harmonic oscillator in which the motion is damped by the action of an additional force is called a damped harmonic oscillator. Damped oscillations are oscillations under the action of resistive forces.

Damping force is $\propto$ to its velocity.

$$\text{ie Damping force} = b \frac{dx}{dt}$$

'b' is called the damping constant. Including the damping force, the differential eq. for a damped harmonic oscillator is

$$m \frac{d^2 x}{dt^2} = -kx - b \frac{dx}{dt}$$

$$m \frac{d^2 x}{dt^2} + kx + b \frac{dx}{dt} = 0$$

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0 \text{ ----- (1)}$$

$$\text{Let } \frac{b}{m} = 2\gamma \text{ where } \gamma \text{ is the damping coefft.}$$

$$\frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega^2 x = 0 \text{ ----- (2)}$$

This is the differential eqn for damped harmonic oscillator

Solutions to the Equation

Assume the solution is of the form

$$x = Ae^{\alpha t}$$

$$\frac{dx}{dt} = \alpha Ae^{\alpha t}$$

$$\frac{d^2x}{dt^2} = \alpha^2 Ae^{\alpha t} = \alpha^2 x$$

Putting these values in (2), we get

$$\alpha^2 x + 2\gamma\alpha x + \omega^2 x = 0$$

$$\text{ie } \alpha^2 + 2\gamma\alpha + \omega^2 = 0$$

$$\text{The solution } \alpha = \frac{-2\gamma \pm \sqrt{(4\gamma^2 - 4\omega^2)}}{2}$$

$$x = Ae^{(-\gamma \pm \sqrt{(\gamma^2 - \omega^2)})t}$$

The solutions can be written as

$$x_1 = A_1 e^{(-\gamma + \sqrt{(\gamma^2 - \omega^2)})t} \text{ and } x_2 = A_2 e^{(-\gamma - \sqrt{(\gamma^2 - \omega^2)})t}$$

$\therefore$ The general solution is $x = x_1 + x_2$

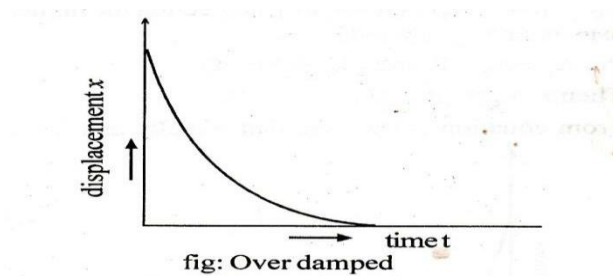
$$x = A_1 e^{(-\gamma + \sqrt{(\gamma^2 - \omega^2)})t} + A_2 e^{(-\gamma - \sqrt{(\gamma^2 - \omega^2)})t} \text{ --- (3)}$$

Where A_1 and A_2 are constants which depends on the initial values of position and velocity.

' γ ' determines the behaviour of the system.

Displacement curve

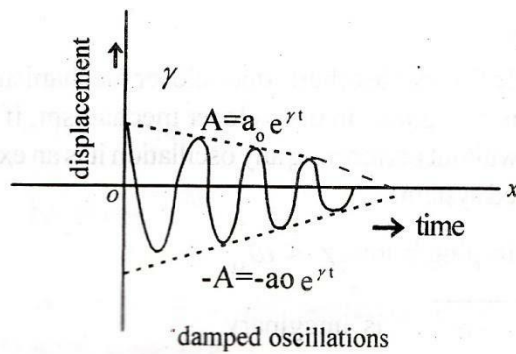
a) overdamped



b) Critically damped



c) under damped

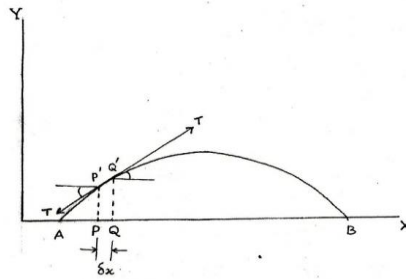


(b)

$$\begin{aligned}
 B &= 8 \times 10^4 \\
 \omega &= 200 \text{ Hz} \\
 B &= \omega Z \\
 Z &= \frac{B}{\omega} = \frac{B}{2\pi\nu} \\
 &= \frac{8 \times 10^4}{2 \times \pi \times 200} \\
 &= 63.69 \text{ sec} \\
 \text{Energy } E &= E_0 e^{-t/Z} \\
 \text{Let } t &\text{ be the time in which energy reduces to } \frac{1}{10} \text{ of the initial value, then} \\
 \frac{E_0}{10} &= E_0 e^{-t/Z} \\
 \frac{1}{10} &= e^{-t/Z} \\
 10 &= e^{t/Z} \\
 \log 10 &= t/Z \\
 t &= Z \log_e 10 \\
 &= 63.69 \times 2.303 \\
 &= 146.67 \text{ sec}
 \end{aligned}$$

12 (a) Consider an infinitely long thin and uniform string stretched between two points by a constant tension T along the X -axis. Let the string be slightly displaced aside along the Y -axis and released. Transverse vibrations are set up in the string.

Consider a small element PQ of the string of length δx . The magnitude of tension in the string will be same everywhere since the string is perfectly flexible. The tension T acts tangentially at every point on the string. $P'Q'$ is the displaced position of the string.



The tension at P' and Q' acts along the tangent making angles θ_1 and θ_2 with the horizontal. Resolving the tension along x- axis and y- axis,

Then net force on PQ acting X and Y direction are

$$F_x = T \cos \theta_2 - T \cos \theta_1$$

$$F_y = T \sin \theta_2 - T \sin \theta_1$$

For small oscillations θ_1 and θ_2 are very small.

$$\cos \theta_1 = 1 \quad ; \quad \cos \theta_2 = 1$$

$$\text{Also } \sin \theta_1 = \tan \theta_1 \quad ; \quad \sin \theta_2 = \tan \theta_2 \quad (\tan \theta_1 = \sin \theta_1 / \cos \theta_1 = \sin \theta_1)$$

$$\text{Then } F_x = 0$$

$$F_y = T \tan \theta_2 - T \tan \theta_1$$

So the net force acting on the element δx in the displaced position is only along Y-axis.

$$F_y = T(\tan \theta_2 - \tan \theta_1)$$

If μ is the mass per unit length of the string (linear density) ,

mass of the element δx , $m = \mu \delta x$

$$\text{Acceleration} = \frac{\partial^2 y}{\partial t^2}$$

According to the Newton's second law of motion

$$F = ma$$

$$F = \mu \delta x \frac{\partial^2 y}{\partial t^2}$$

$$T(\tan \theta_2 - \tan \theta_1) = \mu \delta x \cdot \frac{\partial^2 y}{\partial t^2}$$

$$\text{But } \tan \theta_1 = \left(\frac{\partial y}{\partial x} \right)_x \quad ; \quad \tan \theta_2 = \left(\frac{\partial y}{\partial x} \right)_{x+\delta x}$$

$$\text{or } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\tan \theta_2 - \tan \theta_1}{\delta x} \right]$$

$$\text{ie } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right]$$

when $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right] = \frac{\partial^2 y}{\partial x^2} \text{---(5)}$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} \text{-----(6)}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2}$$

This is the differential equation of a vibrating string.

Comparing this equation with the standard wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

$$\frac{1}{v^2} = \frac{\mu}{T}$$

$$v^2 = \frac{T}{\mu}$$

$$v = \sqrt{\frac{T}{\mu}}$$

This is the velocity of transverse wave on a stretched string.

$$v = \nu \lambda$$

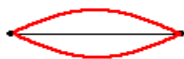
$$\nu = \frac{v}{\lambda}$$

$$\nu = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$$

This is the frequency of transverse waves developed on a stretched string.

NOTE :

If the string is 'l' meters long and vibrating in one segment only, then $l = \frac{\lambda}{2}$



$$\lambda = 2l$$

The frequency of transverse wave is given by

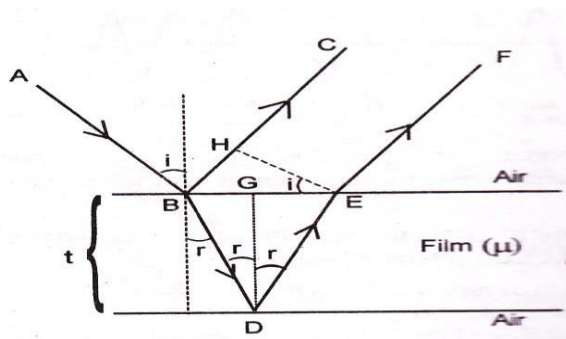
$$v = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

(b)

$$\begin{aligned} T &= mg \\ &= 4 \times 9.8 \\ \mu &= \frac{m}{l} = \frac{2 \times 10^{-3}}{1} \\ v &= \frac{1}{2 \times 1} \sqrt{\frac{4 \times 9.8}{2 \times 10^{-3}}} \\ &= \underline{\underline{70 \text{ Hz}}} \end{aligned}$$

13.(a) When a beam of light falls on a thin film, apart of light is reflected from the top surface of the film and a part is reflected from the lower surface of the film. These two reflected rays interfere. If the incident light is white, the film is beautifully coloured.

Consider a thin transparent film of thickness t and refractive index μ . A ray of light AB is incident on the upper surface of the film. A part of the ray is reflected along BC. The remaining part is transmitted along BD.



Since the film is very thin the rays BC and EF are very close to each other hence they interfere and produce brightness or darkness according to the path difference between them. Draw EH perpendicular to BC. Then beyond EH, the two reflected rays travel the same distance.

The optical path difference = (BD + DE) in film – BH in air

$$\text{o.p.d} = (BD + DE) \mu - BH$$

ΔBDG and ΔEDG ARE congruent

$$BD = DE \text{ and } BG = GE$$

$$\text{o.p.d} = 2 BD \mu - BH \dots\dots\dots(1)$$

From ΔBEH

$$\sin i = BH/BE$$

$$BH = BE \sin i$$

$$BE = BG + GE \text{ and } BG = GE$$

$$BH = 2 BG \sin i$$

Substitute in eqn (1)

$$\text{o.p.d} = 2 BD \mu - 2 BG \sin i \dots\dots\dots(2)$$

From ΔBDG

$$\cos r = DG/BD = t/BD$$

$$BD = \frac{t}{\cos r} \dots\dots\dots(3)$$

$$\tan r = BG/t$$

$$BG = t \tan r \dots\dots\dots(4)$$

Substitute in eqn (3) and (4) in eqn (2)

$$\begin{aligned} \text{Opd} &= \frac{2\mu t}{\cos r} - 2t \tan r \sin i \\ &= \frac{2\mu t}{\cos r} - 2t \frac{\sin r}{\cos r} \sin i \\ &= \frac{2\mu t}{\cos r} (1 - \sin^2 r) \\ &= 2\mu t \cos r \end{aligned}$$

The reflection at B is at the surface of a denser medium. Hence reflected ray BC undergoes a phase change of π or the ray travels an additional distance of $\lambda/2$

$$\text{Correct path difference} = 2\mu t \cos r - \lambda/2$$

CONDITION FOR CONSTRUCTIVE INTERFERENCE

$$\text{Path difference} = n\lambda$$

$$2 \mu t \cos r - \lambda/2 = n\lambda$$

$$2 \mu t \cos r = (2n+1) \lambda/2$$

The film appears bright

CONDITION FOR DESTRUCTIVE INTERFERENCE

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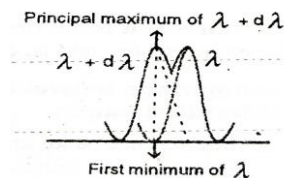
The film appears dark

(b)

$$\begin{aligned} r_n &= 0.162 \text{ cm} \\ r_{n+20} &= 0.368 \text{ cm} \\ D_n &= 0.324 \text{ cm} \\ D_{n+20}^2 &= 0.736 \text{ cm} \\ D_{n+20}^2 - D_n^2 &= 4KR\lambda \\ R &= \frac{D_{n+20}^2 - D_n^2}{4K\lambda} = \frac{(0.736 \times 10^{-2})^2 - (0.324 \times 10^{-2})^2 \text{ m}^2}{4 \times 20 \times 546 \times 10^{-9} \text{ m}} \\ &= \underline{\underline{0.999 \text{ m}}} \end{aligned}$$

14. (a) RAYLEIGH'S CRITERION FOR RESOLUTION OF SPECTRAL LINES

According to Rayleigh's criterion for resolution, two neighbouring spectral lines will be just resolved when the principal maximum of one in any order falls on the first minimum of the other in the same order

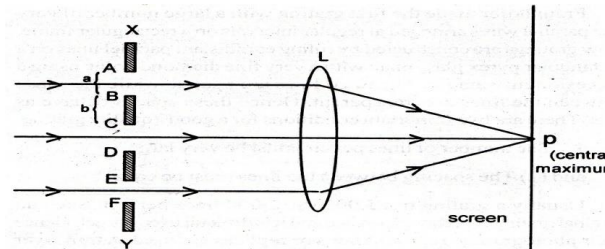


Let λ and $\lambda + d\lambda$ be the wavelengths corresponding to two neighbouring spectral lines of the same order. Then, two spectral lines are visible as separate when the principal maximum of wavelength $\lambda + d\lambda$ falls on the first minimum of wavelength λ of the same order.

PLANE TRANSMISSION GRATING is a plane glass plate containing a large number of equidistant parallel lines drawn using a fine diamond point . The space between the lines acts as narrow slits through which light is transmitted. The lines are opaque to light.

Grating is an arrangement of a large number of parallel slits of equal width separated by equal opaque spaces . Usually a grating has 5000 to 12000 lines per cm. There are two important conditions for a good quality grating.

1. The number of lines per cm must be very large.
2. The spacing between the lines must be equal.

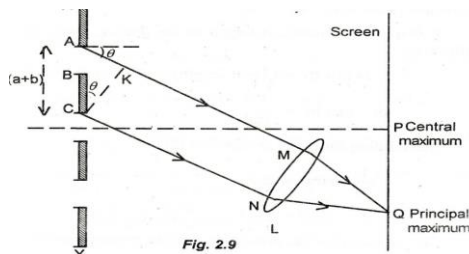


Consider a plane transmission grating placed perpendicular to the plane of paper. AB represents a slit and BC represents a line. Let 'a' be the width of each slit and 'b' be the width of each line. The distant (a+b) is called grating element or grating constant.

A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit. Each point of the wave front sends out secondary waves in all direction. The straight and parallel waves from each point can be focussed on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called central maximum.

The position of central maximum is same for all the wavelength. The central maxima will have the same colour as the incident light.

Consider two waves diffracted from two points A and C of slit. They travel along AM and CN. Draw CK perpendicular to AM. Then the path difference between the two waves is AK.



From triangle ACK

$$\sin \theta = AK/AC$$

$$AK = AC \sin \theta$$

$$AK = (a+b) \sin \theta$$

$$\text{If } (a+b) \sin \theta = n\lambda \text{(1)}$$

where $n=0,1,2,3,\dots$ two waves interfere constructively. This is called principle maximum. For different values of n , there are different values of θ .

If $n=1$, it is the first order principle maximum, if $n=2$, it is the second order principal maximum and so on. Thus on either side of central maximum, a number of principal maxima are obtained.

If there are N lines/unit length of the grating, there are N slit also.

$$N(a+b) = 1 \text{ (unit length)}$$

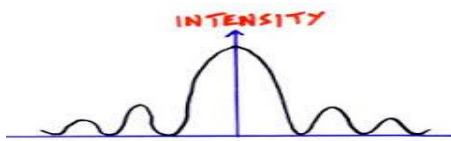
$$(a+b) = 1/N$$

Substitute in equation (1)

$$1/N \sin\theta = n\lambda$$

$$\sin\theta = Nn\lambda$$

This is known as grating law or grating equation.



14.(b)

$$\begin{aligned} \sin\theta &= Nn\lambda \\ \theta &= 30^\circ \\ n &= 2 \\ \lambda &= 520 \text{ nm} = 520 \times 10^{-9} \text{ m} \\ N &= \frac{\sin\theta}{n\lambda} \\ &= \frac{\sin 30}{2 \times 520 \times 10^{-9}} \\ &= 0.000480769230 \times 10^9 \\ &= 480769 \\ &= \underline{\underline{480.8 \times 10^3 \text{ lines/m}}} \end{aligned}$$

15. (a) TIME DEPENDENT SCHRÖDINGER EQUATION FOR A FREE PARTICLE

Schrodinger's equation is the basic expression in quantum mechanics. We can derive it by considering the plane wave equation and combining with Einstein's eqn for quantum energy and de Broglie expression for wavelength.

A particle in motion is associated with a wave function that contains the information about the motion.

A plane wave that propagates along x- direction is given by

$$\Psi = Ae^{i(kx - \omega t)}$$

where k is the wave vector, $k = 2\pi/\lambda$

ω is the angular frequency, $\omega = 2\pi\nu$

Einstein's formula for photon energy is

$$E = h\nu = \frac{h\omega}{2\pi} = \hbar\omega; \text{ where } \hbar = \frac{h}{2\pi}$$

de-Broglie's expression for matter wavelength is $\lambda = \frac{h}{p}$

$$\therefore p = \frac{h}{\lambda} = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} = \hbar k$$

Using the above expression, equation (1) becomes

$$\Psi = Ae^{\frac{i}{\hbar}(\hbar kx - \hbar\omega t)}$$

$$\text{ie } \Psi = Ae^{\frac{i}{\hbar}(px - Et)}$$

On partial differentiation of Ψ with respect to x, twice, we get

$$\frac{\partial^2 \Psi}{\partial x^2} = \frac{-p^2}{\hbar^2} \Psi \dots \dots \dots (2) \quad \text{and}$$

Differentiating with respect to time,

$$\frac{\partial \Psi}{\partial t} = \frac{-iE}{\hbar} \Psi \dots \dots \dots (3)$$

$$p^2 \Psi = -\hbar^2 \frac{\partial^2 \Psi}{\partial x^2} \dots\dots\dots(4)$$

$$= -i\hbar \frac{\partial}{\partial x} \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi \dots\dots\dots(4a)$$

From 3 and 4, $p = -i\hbar \frac{\partial}{\partial x}$

This is called momentum operator:

$$E \Psi = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(5)$$

$$\therefore E = i\hbar \frac{\partial}{\partial t} \dots\dots\dots(5a)$$

This is called Energy operator.

For a free particle total energy is given by

$$E = \frac{p^2}{2m}, \text{ since } V = 0$$

$$\text{ie } E \Psi = \frac{p^2}{2m} \Psi \dots\dots\dots(6)$$

Using equation (4) and (5), Equation (6) becomes

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(7)$$

This is Schrodinger's equation for a free particle in one dimension.

In three dimension it becomes for a free particle,

$$\frac{-\hbar^2}{2m} \nabla^2 \Psi = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(8)$$

$$\text{where } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

is called *Laplacian operator*

If the particle is moving under a potential $V(r,t)$ then equation (8) becomes

$$\left[\frac{-\hbar^2}{2m} \nabla^2 + V_{(r,t)} \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

This is Schrodinger's time dependent equation

TIME INDEPENDENT SCHRÖDINGER EQUATION (STEADY STATE EQUATION)

According to Schrodinger, de- Broglie's wavelength holds good for any particle moving in any field of force with potential energy V .

Then total energy $E = \text{Kinetic energy} + \text{Potential Energy}$

$$E = \frac{1}{2} m v^2 + V$$

$$= \frac{p^2}{2m} + V$$

$$\text{ie } p^2 = 2m(E - V)$$

$$\text{or } p = [2m(E - V)]^{1/2}$$

The wave equation in Cartesian co-ordinate system can be written as

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{u^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots(9)$$

$$\text{where } \Psi = \Psi_0 e^{-i\omega t} \dots\dots\dots(10)$$

' u ' is the velocity of motion of wave and $\Psi(x, y, z, t)$ represents the amplitude of the wave associated with the particle

From (10), we get

$$\frac{\partial^2 \Psi}{\partial t^2} = -\omega^2 \Psi$$

So Equation (9) becomes

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-\omega^2}{u^2} \Psi \dots\dots\dots(11)$$

$$\frac{\omega^2}{u^2} = \frac{(2\pi\nu)^2}{(\nu\lambda)^2} = \frac{4\pi^2}{\lambda^2}$$

So we have

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-4\pi^2}{\lambda^2} \Psi \quad \text{or}$$

$$\nabla^2 \Psi + \frac{4\pi^2}{\lambda^2} \Psi = 0 \dots\dots\dots(12)$$

Equation (12) is a general equation which is independent of time.

Let us now introduce the concept of de-Broglie wavelength.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E - V)}}$$

$$\lambda^2 = \frac{h^2}{2m(E - V)}$$

Substituting the wavelength in eqn (12)

$$\nabla^2 \Psi + \frac{4\pi^2}{h^2} [2m(E - V)] \Psi = 0$$

or

$$\nabla^2 \Psi + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$$

or

$$\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \dots \dots \dots (13)$$

This represents Schrodinger's time - independent wave equation.

For the case of a free particle (V=0), the Schrodinger equation becomes

$$\nabla^2 \Psi + \frac{2mE}{\hbar^2} \Psi = 0$$

(b)

Handwritten calculation for the de Broglie wavelength of an electron with 15 eV energy:

$$\lambda = 546 \text{ nm} = 546 \times 10^{-9} \text{ m}$$

$$E = 15 \text{ eV}$$

$$E = 15 \times 1.6 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 15 \times 1.6 \times 10^{-19}}}$$

$$= 0.317 \text{ nm}$$

16.(a) Electrical properties

- Ionization potential at nano sizes are higher than that at bulk solids.
- The change in electrical properties is not the same in all nano materials.
- In nano ceramics the electrical conductivity increases with the decrease in particle size.
- But in metals it decreases.
- In the case of bulk metals the energy level of electron is continuous whereas in nano sized metals the energy levels are discrete.
- Optical properties
 - Depending on the particle size, same substance shows different colours.
 - Gold spheres of 100 nm size appears orange while that of 50 nm size appears green

- The particles can be made to emit or absorb specific wavelength of light by varying its size

Mechanical properties

- Nano crystalline nickel is as strong as hardened steel. Because of nano size many mechanical properties such as hardness, elastic modulus, fracture toughness, scratch resistance and fatigue strength are modified
- 30-50 % lower elastic moduli than conventional materials
- 2-7 times higher hardness
- Super plastic behaviour in brittle ceramics

Nanomedicine

- Cell imaging
- Treating cancer
- For diagnostic purposes
- The drug consumption and side effects can be reduced significantly by targeted medicine.
- This is done depositing the medicine in the affected region only and in the correct dose
- NT helps to reproduce or repair damaged tissue.

(b) Nano material have a relatively larger surface area to volume ratio when compared to bulk solids. Consider a sphere of radius 'r' .

The surface area $= 4\pi r^2$

Its volume $= \frac{4}{3}\pi r^3$

$$\text{Ratio} = \frac{\text{surface area}}{\text{volume}} = \frac{3}{r}$$

Thus when the radius of the sphere decreases, its surface area to volume ratio increases. Hence as the particle size decreases a greater number of atoms are found at the surface compared to those inside. Thus nano material have more surface area per given volume compared with larger particles. It makes material more chemically reactive as chemical reaction occurs at surfaces.

17.(a) PARAMAGNETISM

- Tendency to move from weak to strong magnetic field when placed in a non uniform magnetic field.
- Unpaired electrons are there.
- Induced field will be in the direction of applied field;but weak
- Free electron magnetism is weak and is called pauliParamagnetism
- They do not retain any magnetism in the absence of an external magnetic field.
- Examples: O₂,Ne,

Curie's Law

$$\chi = \frac{C}{T}$$

FERROMAGNETISM

- Domain concept –In a domain very large number of atoms are aligned parallel to each other.
- Before magnetisation the domains are randomly oriented relative to each other.
- Magnetism persists even after the removal of magnetic field.
- Strongly attracted by external field
- Loss their magnetic properties if they are heated to high temperatures.
- They have high susceptibility and relatively high permeability
- examples-Iron, Cobalt, Nickel

(b)

$\chi_1 = 3.756 \times 10^{-4}$
 $T_1 = 200K$
 $\chi_2 = ?$
 $T_2 = 600K$
 Curies law,
 $\chi = \frac{C}{T}$
 $\chi_1 = \frac{C}{T_1} = \frac{C}{200}$
 $C = 3.756 \times 10^{-4} \times 200$
 $\chi_2 = \frac{C}{T_2} = \frac{3.756 \times 10^{-4} \times 200}{600} = 1.252 \times 10^{-4}$

18(a)

Propagation of Electromagnetic waves in free space

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

In vacuum, $\mathbf{J}=0$, $\rho=0$

Hence the Maxwells equations becomes

$$\nabla \cdot \mathbf{D} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$$

Again in free space $\mathbf{D} = \epsilon_0 \mathbf{E}$, $\mathbf{B} = \mu_0 \mathbf{H}$

$$\nabla \cdot \mathbf{E} = 0 \dots\dots\dots(1)$$

$$\nabla \cdot \mathbf{H} = 0 \dots\dots\dots(2)$$

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \dots\dots\dots(3)$$

$$\nabla \times \mathbf{H} = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \dots\dots\dots(4)$$

Taking curl on both sides of eqn (3)

$$\nabla \times \nabla \times \mathbf{E} = -\mu_0 \nabla \times \frac{\partial \mathbf{H}}{\partial t} \dots\dots\dots(5)$$

In vector Calculus $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$

$$\text{Eqn (5) becomes } \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \nabla \times \frac{\partial \mathbf{H}}{\partial t}$$

$$-\nabla^2 \mathbf{E} = -\mu_0 \frac{\partial}{\partial t} \nabla \times \mathbf{H}$$

$$\nabla^2 \mathbf{E} = \mu_0 \frac{\partial}{\partial t} \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \dots\dots\dots(6)$$

Taking curl on both sides of eqn (4)

$$\nabla \times \nabla \times \mathbf{H} = \epsilon_0 \nabla \times \frac{\partial \mathbf{E}}{\partial t}$$

$$\nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H} = \epsilon_0 \frac{\partial}{\partial t} \nabla \times \mathbf{E}$$

$$-\nabla^2 \mathbf{H} = -\epsilon_0 \mu_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{H}}{\partial t^2} \dots\dots\dots(7)$$

Comparing equations 6 and 7 with the std eqn of wave

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$\frac{1}{v^2} = \epsilon_0 \mu_0$$

$$v = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$v = 3 \times 10^8 \text{ m/s}$$

Thus electromagnetic waves travel in freespace with the velocity of light.

(b)

Gauss theorem (Relation between surface integral and volume integral)

For a closed surface S , surrounding the volume V , the divergence of any vector A is expressed as

$$\int_S \vec{A} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{A}) dV$$



Where $d\vec{s} = ds \cdot \hat{n}$, ds an element of area S and $\hat{n}$ is the unit outward normal to S .

Stoke's theorem



If C is the closed contour around the edge of an open surface S , then Stokes's theorem states that

$$\int_C \vec{A} \cdot d\vec{l} = \int_S (\nabla \times \vec{A}) \cdot d\vec{s}$$

19. (a) all the magnetic lines have been expelled from the interior of the specimen when it is cooled below the transition temperature in the magnetic field. The property of expulsion of

magnetic flux density from the interior of a superconducting material during the transition from normal state to superconducting state is called Meissner effect.

When a specimen is placed in a magnetic field H , magnetic lines are passing through it.

$$B = \mu_0(H + M)$$

Where,

B = flux density inside the specimen

μ_0 = Permeability

M = magnetization

When the specimen is cooled below the transition temperature, all magnetic lines are cancelled from the specimen.

Since $B=0$,

$$H = -M$$

Magnetic susceptibility $\chi = M/H = -1$

Hence the superconductor becomes a perfect diamagnet at the transition temperature.

Type 1 Super conductor

When a superconductor is placed in a magnetic field H , the intensity of magnetization M is induced in it. When H increases, M also increases. The variation of M with H is shown in fig.

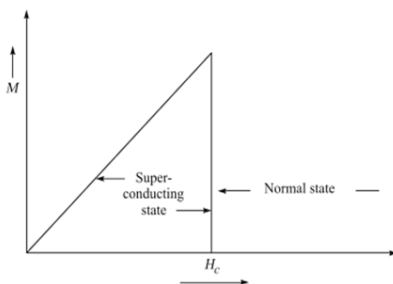


Fig. 19.5 Type I superconductor

Up to the critical field H_c , magnetization increases proportional to the applied field and it abruptly drops to zero at the critical field. Above H_c it behaves as normal conductor. Below H_c , it is in the superconducting state. Up to H_c , it behaves like a diamagnet and strictly obeys Meissner effect. Type of superconductors for which M becomes zero abruptly when $H=H_c$ are called Type 1 superconductors. The transition from superconducting state to normal state is very abrupt. The

critical field H_c is very small in the order of 0.1 or 0.2 Tesla. Therefore it is easy to change type I superconductor into a normal conductor. Hence they are also known as soft superconductors. The transition is reversible.

Eg: Al, Pb, Indium and almost all pure metals.

Type 2 superconductors

In this type there are two critical fields, Lower critical field H_{c1} and upper

critical fields H_{c2} . Up to H_{c1} magnetism is proportional to the applied field and it behaves like Type 1 superconductors. It strictly obeys Meissner effect up to H_{c1} . In between H_{c1} and H_{c2} magnetization decreases gradually. At H_{c2} , the magnetisation vanishes completely and external field penetrates, completely destroying the superconductivity. Above H_{c2} material behaves as normal conductor. Between H_{c1} and H_{c2} material is in a mixed state (it behaves partially conducting as well as partially superconducting) known as vortex state. It does not strictly obey Meissner effect in this region. Transition from superconducting state to normal state is very slow and gradual. The field H_{c2} value is very high in the order of 10T to 20T. A strong magnetic field is required to convert it into a normal conductor. They are known as hard superconductors. The transition is irreversible. Very strong magnetic field can be produced using type II superconductors.

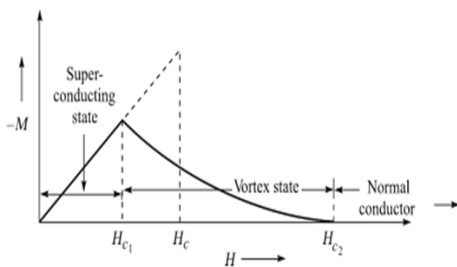


Fig. 19.6 Type II superconductor

Eg: Niobium, Germanium and all alloys

(b) High Temperature Superconductors are ceramic superconductors with a high transition temperature greater than 40 K. In low temperature superconductors it is very difficult and expensive to maintain low temperature for a very long period. It is very difficult to maintain the liquid helium temperature (4.2K). nowadays it is possible to replace the expensive liquid He with cheaper and more efficient coolant liquid nitrogen. The temperature T_c can be raised using microwaves or laser radiations. In ceramic superconductors, the cell contains 1 atom of rare earth metal, 2

Barium atoms, 3 copper atoms and 7 Oxygen atoms. Since no. of atoms of metal are in the ratio 1:2:3, such ceramic superconductors are called 1-2-3 superconductors and it is working at T_c of 90K. Copper oxide superconductors belong to another group of HTSC working at T_c of 120K. The Resonating Valence Bond State Theory explained.

20.(a)

NUMERICAL APERTURE (NA)

Numerical aperture NA of an optic fibre is the sine of acceptance angle θ_a .

$$NA = \sin \theta_a$$

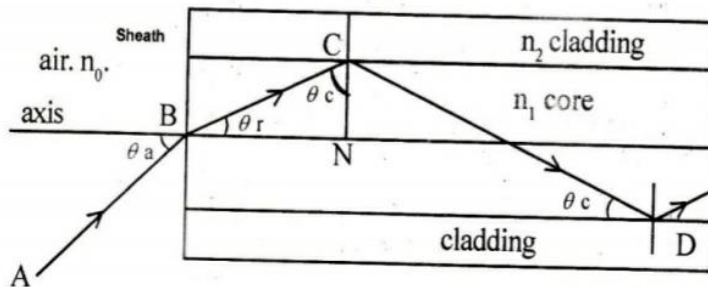
NA depends on the acceptance angle. If θ_a is larger, NA will also be higher.

Numerical aperture is also the light gathering power of the fibre and it measures the amount of light accepted by the fibre. NA is between 0.13 and 0.5. If NA is larger, fibre can accept more light from the source. NA depends on the refractive indices of core and cladding.

$$NA = \sqrt{n_1^2 - n_2^2}$$

PROOF

Consider a light ray AB incident at B at the edge of the core of an optic fibre from air. It is incident at an angle θ_a with the axis of the core. This maximum angle is known as acceptance angle. Since it travels from air to core, it is refracted along BC at an angle θ_r . This refracted ray is now incident at C at the core-cladding interface with an angle slightly greater than the critical angle θ_c . Hence the ray is undergoing total internal reflection and it is travelling along CD. Thus the ray is propagated through the fibre by multiple total internal reflections.



At C, CN is drawn normal to the axis. The angle at C is taken as the limiting angle θ_c , the critical angle. Let n_0 be the refractive index of air, n_1 that of core and n_2 that of cladding.

By Snell's law at B,
$$\frac{\sin \theta_a}{\sin \theta_r} = \frac{n_1}{n_0}$$

$$n_0 \sin \theta_a = n_1 \sin \theta_r$$

$$n_0 \text{ for air} = 1 \text{ and } NA = \sin \theta_a$$

$$NA = n_1 \sin \theta_r \dots\dots\dots(1)$$

At the critical angle,
$$\frac{\sin \theta_c}{\sin 90} = \frac{n_2}{n_1}$$

$$\sin \theta_c = \frac{n_2}{n_1} \dots \dots \dots (2)$$

$$\text{But } \sin \theta_c = \cos (90 - \theta_c)$$

$$\theta_c + \theta_r = 90^\circ$$

$$\sin \theta_c = \cos \theta_r$$

$$\text{From (2) } \cos \theta_r = \frac{n_2}{n_1}$$

$$\text{But } \cos^2 \theta_r + \sin^2 \theta_r = 1$$

$$\sin^2 \theta_r = 1 - \cos^2 \theta_r$$

$$\sin^2 \theta_r = 1 - \frac{n_2^2}{n_1^2}$$

$$\sin^2 \theta_r = \frac{n_1^2 - n_2^2}{n_1^2}$$

$$\sin \theta_r = \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

Substituting for $\sin \theta_r$ in eqn (1)

$$NA = n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$\text{Numerical Aperture, } NA = \sqrt{n_1^2 - n_2^2}$$

Where,

n_1 = Refractive index of core

n_2 = Refractive index of cladding

(b)

$$n_1 = 1.62$$

$$n_2 = 1.52$$

$$N.A = \sqrt{n_1^2 - n_2^2} = \sqrt{1.62^2 - 1.52^2} = \sqrt{0.314}$$

$$= \underline{\underline{0.56035}}$$

$$N.A = \sin \theta_a.$$

$$\theta_a = \sin^{-1}(N.A)$$

$$= \sin^{-1}(0.56035)$$

$$= \underline{\underline{34.1^\circ}}$$